SETP4583

SURFACTANT AND DETERGENT TECHNOLOGY

SECTION EB01

GROUP ASSIGNMENT:

DESIGN AND PROTOTYPE OF AN INNOVATIVE SOAP/ DETERGENT PRODUCT

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1.0 Introduction and Product Concept

PURELUX Gentle Body Wash is a new kind of detergent product designed to have a gentle but high-performance cleansing while environmental sustainability. We made this prototype in a laboratory setting to show that personal care products could minimize the harm towards environment. Our main goal of PURELUX was to create a body wash that is gentle for skin, works effectively and safe for the planet.

The special innovation behind the PURELUX production is how we make it using a cold mix process. Most factories use methods that require significant heat to make soap, but we mix our ingredients at room temperature to save electricity and reduce pollution. We use a smart mix of cleaners like SLES for bubbles, Betaine for mildness and Berol to help remove oil. This mixture ensures formula stability and effective grease-removing performance without being too rough on the skin.

PURELUX is specifically targeted toward environmentally conscious consumers looking for biodegradable personal care options as the ingredients could break down easily in water. It is also designed for individuals with sensitive skin who require a pH-balanced formula by maintaining a pH level between 5.5 and 6.0. This pH level matches the skin's natural balance and prevents skin irritation. We kept the recipe simple by using 77% purified water and do not use extra dyes and perfumes that can cause rashes. Sustainability is further reinforced by using refillable pump-bottle design to encourage reuse and significantly reduce plastic waste.

2.0 Ingredient List and Formulation Rationale

Core Material	Percentage
SLES	12%
Betaine	8%
Berol	2%
LABS	1%
Purified Water	77%

Table 1 List of Ingredients Used

The surfactant formulation that was developed during this study includes Sodium laureth Sulfate (SLES) 12%, Cocamidopropyl Betaine 8%, Berol 2%, Linear Alkylbenzene Sulfonate (LABS) 1% and pure water at about 77%. The selection of these ingredients was aimed at creating a balanced surfactant system to have a good cleaning performance, formulation stability and successful production in laboratory experimentation.

The major anionic surfactant used in the formulation is Sodium Laureth Sulfate (SLES). The reason why it was selected is because it has a good level of detergency and high foaming capacity, which is critical in the removal of oils, grease, and particulate dirt. SLES has found extensive application in liquid detergent and soap preparation due to its dependability, affordability and effectiveness in surfactant-based compound.

Cocamidopropyl Betaine is an amphoteric surfactant which supplements the anionice surfactants of the system. Its main use is enhancement of the stability of foams and making them less harsh like the use of strong anionic surfactants like SLES and LABS. Betaine addition improves the overall cleaning efficacy with improved compatibility since it increases mildness and stability of the formulation.

Berol was added as a non-ionic surfactant to increase the wetting and emulsification characteristics. The non-ionic surfactants do not depend on the water hardness and pH differences hence the formulation can be used to achieve the same results with different conditions. The dispersion of the oily soils is also facilitated by Berol which enhances the overall cleaning efficiency and the formulation balance.

The secondary anionic surfactant in the formulation is Linear Alkylbenzene Sulfonate (LABS). LABS improves the performance in the grease-cutting and soil removal, even though at a relatively low concentration. The combination with SLES enhances detergency and reduces the possible instability of formulation.

The main solvent and carrier of the formulation is purified water. It assures of appropriate dissolution and association of the surfactants, it aids in appropriate viscosity, and aids in homogenous blend. Pure water is also used to avoid contamination and provide uniformity in the preparation process.

In general, synergistic mixture of anionic, amphoteric and non-ionic surfactants was used in the formulation. The ratios and concentrations chosen were well thought of in order to compromise between cleaning efficiency, foam stability, mildness, and physical stability. Laboratory tests have demonstrated that this surfactant system was successfully obtained, which confirms that the choice of ingredients and the composition design was suitable to develop a prototype detergent.

3.0 Preparation Method and Process

The preparation of the body wash was carried out by using a simple laboratory-based formulation process with the available core surfactants. The main ingredients used were purified water, SLES, betaine, Berol, and LABS. Due to limited resources during experimentation, additional additives such as glycerin, fragrance, preservatives, and colorants were not consistently available and therefore were not included in the final formulation.

Initially, purified water was placed into a container as the base solvent. SLES in solid form was then added into the water while continuously stirring to ensure complete dissolution. Adequate stirring was required to avoid clumping and to achieve a uniform solution. After SLES was fully dissolved, betaine and Berol were added to enhance foam formation and improve mildness. LABS was then incorporated at a low concentration to strengthen the cleansing performance without making the formulation overly harsh. Mixing was maintained throughout the process to create a well-mixed solution.

During the formulation, adjustments were mainly focused on modifying the percentage of betaine to improve skin mildness. No heating was applied, and all mixing was conducted at room temperature. Syringes were used as measurement tools once they became available, allowing for more accurate control of ingredient ratios.

Figure 1 Slime-like Texture Failed Result

One major challenge faced during this process was the lack of proper measuring equipment at the initial stage. At first, measurements were estimated using bottle caps, which resulted in inconsistent ratios and caused the formulation to develop a slime-like texture rather than a body wash-like consistency as shown in Figure 1 above. After identifying this issue, syringes were introduced as alternative measuring tools. Although slightly difficult to use, it improved accuracy. Following several trials and adjustments, the formulation process became more controlled, leading to a stable body wash product with suitable texture and performance.

4.0 Test Results and Performance Evaluation

The performance of our body wash was evaluated based on visual observation and practical usage tests. The final product demonstrated effective cleaning performance, successfully removing dirt and oil from the skin. Foam formation was satisfactory, means that the surfactant combination functioned as expected. The final formulation produced a body wash with a texture close to that of commercial products as shown in Figure 2 below, showing that the viscosity and consistency were within a suitable range.

Figure 2 Body Wash with Success Sample

In terms of usability, the body wash was easy to apply and rinse off, with no visible residue left on the skin after use. Compared to commercial body wash products, the formulation met basic expectations in terms of cleansing ability and foaming performance. However, a noticeable limitation was observed regarding skin feel after use. The skin tended to feel slightly dry, likely due to the absence of moisturizing additives such as glycerin or conditioning agents.

This issue became more apparent during group presentations, where other groups demonstrated improved formulations by incorporating ingredients like aloe vera to enhance skin hydration. Based on this observation, it was concluded that the dryness issue could be addressed by adding moisturizing or soothing agents in future formulations.

The body wash achieved its primary function of cleansing effectively and producing adequate foam. While the performance was acceptable for a basic formulation, further improvements can be made by enhancing mildness and skin-conditioning properties to better match commercial standards.

5.0 Biodegradables and Toxicity Assessment

The formulation design of PURELUX Gentle Body Wash prioritizes environmental sustainability and consumer safety, aiming to achieve a balance between functional cleaning performance and a minimal ecological footprint. The surfactant system is primarily composed of Sodium Laureth Sulfate (SLES) and Cocamidopropyl Betaine, both selected for their established biodegradability under aerobic conditions. This characteristic ensures that the surfactants degrade rapidly in wastewater treatment systems, thereby preventing long-term bioaccumulation in aquatic environments. While Linear Alkylbenzene Sulfonate (LABS) was utilized for its superior detergency, its concentration was strictly restricted to 1% to mitigate its potential aquatic toxicity risks while retaining its cleaning efficacy. Furthermore, the inclusion of the non-ionic surfactant Berol enhances the emulsification of oils without compromising the formulation's environmentally friendly profile.

In terms of human safety and toxicity, the formulation was engineered to minimize dermatological irritation. Although anionic surfactants like SLES provide effective cleansing, they possess a potential for skin irritation. To counteract this, 8% Cocamidopropyl Betaine was incorporated as a co-surfactant. Its synergistic interaction with SLES significantly reduces the harshness of the formulation, enhancing overall skin mildness. The prototype further minimizes toxicity risks by adopting a hypoallergenic formulation strategy, excluding non-essential synthetic dyes, heavy fragrances, and parabens to lower the incidence of contact dermatitis. Additionally, the product maintains a pH range of 5.5 to 6.0, which aligns with the physiological pH of the human skin mantle, ensuring that the product cleanses effectively without disrupting the skin’s natural moisture barrier.

6.0 Sustainability and Safety Considerations

Our project focused on being eco-friendly in how we made the soap and how it is used. The biggest benefit is our "cold mix" process. Unlike many factories that use a lot of heat to make soap, we mixed our ingredients at room temperature. This saves a lot of electricity and reduces our carbon footprint. We also kept the recipe simple by using only water (77%) and necessary cleaners. We did not add extra chemicals just to make the soap look thick or shiny, which means fewer chemicals go down the drain. The soap also rinses off quickly, which helps people use less water in the shower.

To keep users safe, our packaging has clear instructions. The label warns users to keep the soap away from their eyes and says it is for external use only. For packaging, we recommend using a bottle with a pump. This design is easy to refill, which encourages people to reuse the bottle instead of throwing it away. This supports our goal of reducing plastic waste and keeping the product sustainable.

7.0 Reflection on Team Collaboration and Learning

Figure 3 Group Photo

Our team, consisting of Poh Lok Yee, Chau Ying Jia, Gui Kah Sin, Lau Yee Wen and Woo Cheng Shuan worked together to develop the PURELUX Gentle Body Wash prototype. We collaborated by following a laboratory-based process where we shared the tasks of measuring, stirring, and testing the ingredients. During the project, we faced a major challenge when we did not have proper measuring tools at the start. We tried to estimate the amounts of surfactants using bottle caps, which caused the mixture to fail and turn into a slime-like texture instead of a smooth body wash. We solved this problem by switching to syringes to measure the ingredients more accurately which eventually led to a successful and stable product.

Through this process, we learned several important skills, including technical skills like balancing surfactants and research skills to find biodegradable ingredients. We also improved our communication skills by discussing our observations and preparing our group presentation together. We gained valuable insights into surfactant technology, specifically learning that mixing ingredients at room temperature, a cold mix process saves a lot of energy and reduces the carbon expose. We also realized how important it is to balance ingredients like SLES with Betaine to make sure the soap is effective but not too harsh on the skin.

For future teamwork, we identified a few areas that could be improved to make our product better. While our body wash cleaned effectively, it made the skin feel slightly dry because we lacked moisturizing ingredients like glycerin and aloe vera. After watching other groups' presentations, we saw that adding aloe vera or other soothing agents could significantly improve skin hydration. In the future, we plan to improve our product by researching additional conditioning agents earlier during the project to ensure the final product meets commercial standards both for comfort and quality.